

Review of Large Language Models for Antibiotic Prescription and Antimicrobial Resistance: Policy-Oriented Analysis

1. Objective

Antimicrobial resistance (AMR) is considered one of the most significant global public health threats. In recent years, large language models (LLMs) such as ChatGPT and similar generative AI systems have begun to appear in medical research and clinical decision-support contexts. These systems are being explored for tasks such as antibiotic prescription guidance, antimicrobial resistance detection, and antimicrobial stewardship support.

The objective of this project was to review the existing literature on the application of large language models in the context of antibiotic prescription and antimicrobial resistance. The review specifically focuses on evaluating these studies from a policy and governance perspective. In particular, the analysis considers how current research aligns with international policy frameworks on AI governance in healthcare, including the **World Health Organization (2021) guidelines on Ethics and Governance of Artificial Intelligence for Health** and the **OECD Recommendation on Artificial Intelligence (2024)**.

The goal of the review is to understand the current state of research, identify limitations in existing studies, and highlight areas where stronger policy guidance or governance mechanisms may be required before these technologies can be safely integrated into healthcare systems.

2. Methods

A literature search was conducted using PubMed to identify research related to large language models and antibiotic prescribing or antimicrobial resistance. The following search query was used:

("large language model" OR ChatGPT OR "generative AI") AND ("antibiotic prescription" OR "antimicrobial stewardship" OR AMR)

This search returned **29 articles**.

The screening process was conducted in two stages:

1. Title and abstract screening:

Articles that were not directly related to LLM applications in antibiotic prescribing, antimicrobial resistance detection, or antimicrobial stewardship were excluded.

2. Full-text screening and data extraction:

Relevant papers were then examined in detail and key information was extracted.

After screening, **21 studies were included** for further review, while **8 articles were excluded** due to lack of relevance.

For each study, the following data were extracted and organized in a structured spreadsheet:

- PMID
- Title
- Authors
- Journal
- Publication year
- Methodology
- Prompts used
- Number of LLMs evaluated
- Comparison with infectious disease specialists or clinical standards
- Key findings
- Limitations
- Future research directions
- Policy or ethical considerations

The final analysis focused particularly on studies that examined **LLMs for antibiotic prescribing, antimicrobial resistance detection, or antimicrobial stewardship support**, and these were interpreted through existing AI governance frameworks.

3. Results

The reviewed literature shows that research on LLMs and antimicrobial resistance currently falls into three broad categories:

1. **Clinical decision support for antibiotic prescribing**
2. **Detection or interpretation of antimicrobial resistance mechanisms**
3. **Public health communication and antimicrobial stewardship support**

A summary of key studies relevant to antibiotic prescribing and AMR is provided below.

Study	Focus	Key Findings	Major Limitations
De Vito et al. (2025)	Comparison of 14 LLMs for antibiotic prescribing	Best-performing model achieved ~71% correct prescriptions; large variability between models	Accuracy declined in complex cases
De Vito et al. (2024)	ChatGPT vs infectious disease specialists	LLMs performed well on theoretical questions but struggled with clinical cases and antibiograms	Tendency to recommend longer treatment durations
Montiel-Romero et al. (2025)	Agreement between ChatGPT and ID specialists	Moderate agreement for antibiotic choice (51%); lower agreement for resistance mechanism identification	Simulated cases, limited real-world validation
Giske et al. (2024)	GPT-4 detection of resistance mechanisms	High sensitivity for detecting resistance mechanisms when trained with guidelines	Lower specificity compared to human microbiologists
Zama et al. (2025)	Untrained LLM in pediatric antibiotic prescribing	Model produced generally acceptable responses in simple pediatric cases	Very small sample (13 scenarios)

Overall, the studies suggest that LLMs demonstrate promising capabilities but inconsistent reliability in antibiotic prescribing tasks. While some models show strong performance in theoretical knowledge or structured scenarios, accuracy decreases in more complex clinical situations, particularly those involving detailed microbiological data.

Another important observation is that model performance improves significantly when clinical guidelines are explicitly incorporated, such as the use of EUCAST antimicrobial susceptibility guidelines in AI systems.

However, none of the reviewed studies concluded that LLMs are ready for fully autonomous clinical prescribing.

4. Discussion

The current literature suggests that LLMs have significant potential in antimicrobial stewardship and clinical decision support. However, several limitations remain that are directly relevant to existing international AI governance frameworks.

One key principle emphasized by both the WHO and OECD guidelines is **human oversight and accountability in AI-assisted healthcare decision-making**. Most studies reviewed in this analysis support this principle. While LLMs can generate antibiotic recommendations, their outputs were inconsistent and sometimes clinically inappropriate. Several studies reported tendencies toward incorrect antibiotic choices, overly long treatment durations, or low accuracy when interpreting microbiological data.

This suggests that LLMs may currently be more appropriate as **support tools for clinicians rather than autonomous decision-makers**. In antimicrobial stewardship programs, these models could assist with literature summarization, guideline retrieval, or analysis of clinical cases, but final prescribing decisions should remain with trained medical professionals.

Another policy concern highlighted by the literature is **validation and safety testing**. Many of the studies rely on simulated clinical cases rather than real-world patient data. Additionally, some research uses small datasets or limited clinical scenarios. According to WHO guidance, AI systems used in healthcare should undergo rigorous validation, including external testing across multiple healthcare settings. The current evidence base does not yet meet this standard.

The issue of **bias and generalizability** is also important. Several studies were conducted using datasets from specific hospitals or geographic regions. This raises questions about whether the same models would perform equally well in other clinical environments with different patient populations, pathogen distributions, or prescribing practices.

A further concern is **transparency and explainability**. LLMs often produce recommendations without clearly explaining the reasoning process behind them. In clinical medicine, this lack of interpretability can create challenges for accountability and trust. Policy frameworks such as the OECD AI principles emphasize the importance of transparency and explainability, particularly for high-risk applications such as medical decision-making.

Finally, governance structures for AI deployment in healthcare are still evolving. Many studies acknowledge the need for **clear regulatory pathways, monitoring systems, and post-deployment evaluation** before LLM-based tools can be integrated into clinical workflows.

5. Future Directions

Future research should focus on improving the reliability and safety of LLM-based clinical decision-support tools for antimicrobial prescribing.

Several important steps are needed:

1. **Larger clinical validation studies**
Research should include real-world patient cases across multiple hospitals and healthcare systems.
 2. **Integration with clinical guidelines**
Models that explicitly incorporate antimicrobial stewardship guidelines appear to perform better and should be further explored.
 3. **Improved interpretability**
Developing methods that allow LLMs to explain their reasoning would help clinicians assess the reliability of AI-generated recommendations.
 4. **Standardized evaluation frameworks**
Currently, different studies use different methods to assess LLM performance. Establishing common evaluation standards would make it easier to compare results across models.
 5. **Policy and governance integration**
AI systems for antibiotic prescribing should be designed in alignment with international governance frameworks such as the WHO and OECD guidelines, particularly regarding safety, accountability, transparency, and human oversight.
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6. Conclusion

The current body of research suggests that large language models have promising applications in antimicrobial stewardship and antibiotic prescribing support. However, their reliability remains inconsistent, especially in complex clinical situations. Most studies emphasize that these systems should be used only as **decision-support tools rather than replacements for clinicians**.

From a policy perspective, the findings reinforce the importance of strong governance frameworks for AI in healthcare. International guidelines from the WHO and OECD highlight key requirements such as human oversight, transparency, safety validation, and accountability. The reviewed literature indicates that while LLM technology is advancing rapidly, significant work remains before these systems can safely be integrated into routine clinical practice.

Continued research, combined with robust regulatory oversight, will be essential to ensure that AI tools contribute positively to global efforts to combat antimicrobial resistance.